♠ BLACKJACK LAB

SESSION REPORT · 001
October 29, 2026

A FLAT-BET SESSION AGAINST THE CPU DEALER

100 hands *against the house.*

Six decks, one unit a hand, and the total-dependent chart.
No bet spread. No hunches. One hundred chances to be right.



NET RESULT

+0.00 units

ROUNDS	W–L–P	OPTIMAL	EV LOST
100	39–42–19	114/114	0.000

6 decks · S17 · 3:2 · Double any 2 · DAS · 3 resplits

BANKROLL +200.00 → +200.00 · SEED 20261029 · COMMIT 6e0fc28

THE SHORT VERSION

A break-even session *is not a break-even player.*

After 100 hands, the bankroll was exactly where it started: 200.00 units in, 200.00 out. Underneath that flat result, the sample was unusually push-heavy — and the decisions were exactly what the chart ordered.

NET RESULT	EXACT EXPECTATION	COACH EV LOST	ACCURACY
+0.00	-0.43	0.000	100.0%
units over 100 rounds	at the 0.426% fresh-shoe edge	114 graded decisions	all graded plays optimal

Scorecard

Rounds	100	Hands	104
Wins	39	Losses	42
Pushes	19	Naturals	4
Dealer BJs	4	Doubles	10
Splits	4	Insurance	6 offers / 1 taken
High	207.00	Low	198.00

Four split rounds produced the 104 hands. The bankroll peaked at 207.00 after hand 17 and bottomed at 198.00 after hand 94.

The line

S	Stand	63
H	Hit	49
D	Double	10
P	Split	4
I	Insurance	6

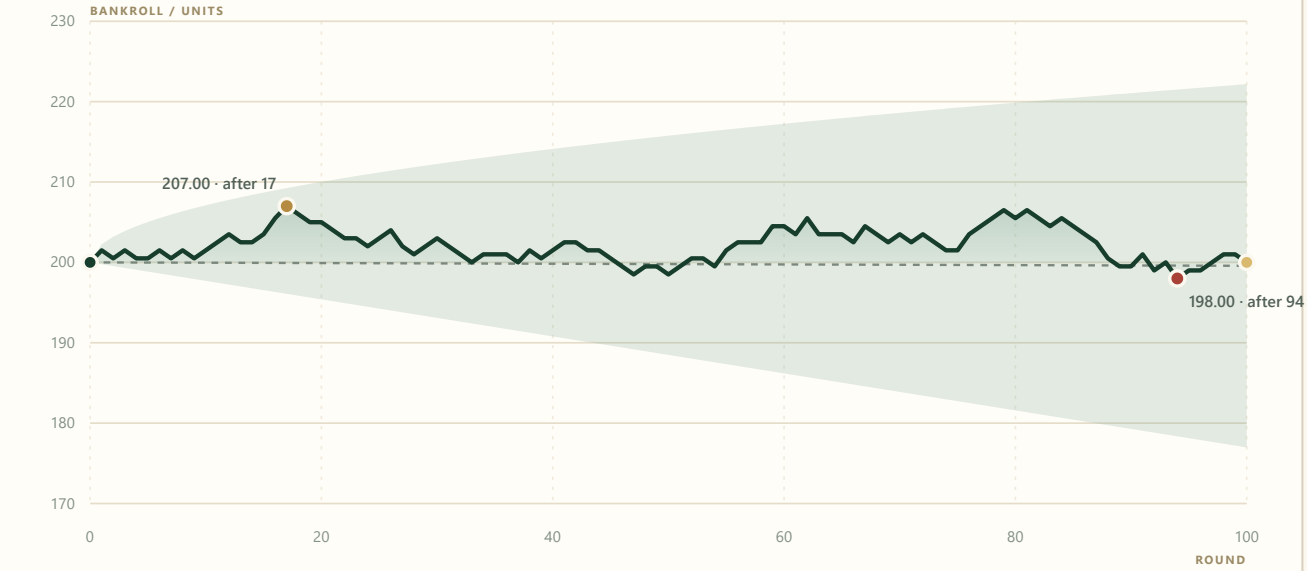
114 of those actions carried a chart grade. All 114 were optimal. The remaining actions were rare post-hit states without a chart cell.

> THE VERDICT
>
> Perfect play and a flat bankroll can coexist. The edge is only 0.43 units per 100 hands; the standard deviation is roughly 11.5. This session did not beat the house. It also did not prove the house was beating it.

THE LUCK CONE

The path stayed *inside the noise.*

A 100-hand sample cannot settle a 0.426% edge. The shaded cone is the 95% range for a flat-betting player with the same policy, starting from the exact expectation. The bankroll line never leaves it.



Expected path: 200.00 → 199.57. The 95% band at hand 100 is roughly ±22.6 units, so even a perfect player's 100-hand result is dominated by variance.

PEAK	LOW	LONGEST WIN STREAK	LONGEST LOSS STREAK
207.00	198.00	4	5
after hand 17	after hand 94	hands in a row	hands in a row

READING THE CHART

The session finished **+0.43 units above** the exact fresh-shoe expectation. That is only **0.04 standard deviations** — indistinguishable from noise. The peak of 207.00 and the trough of 198.00 both fit comfortably inside the cone.

100 HANDS VS 100,000

The sample was *push-heavy.*

The same rules and chart policy played 100,000 rounds. The reference is not a prediction for this session; it is a long-run baseline for the outcome mix. The session's zero-net rounds stand out immediately.

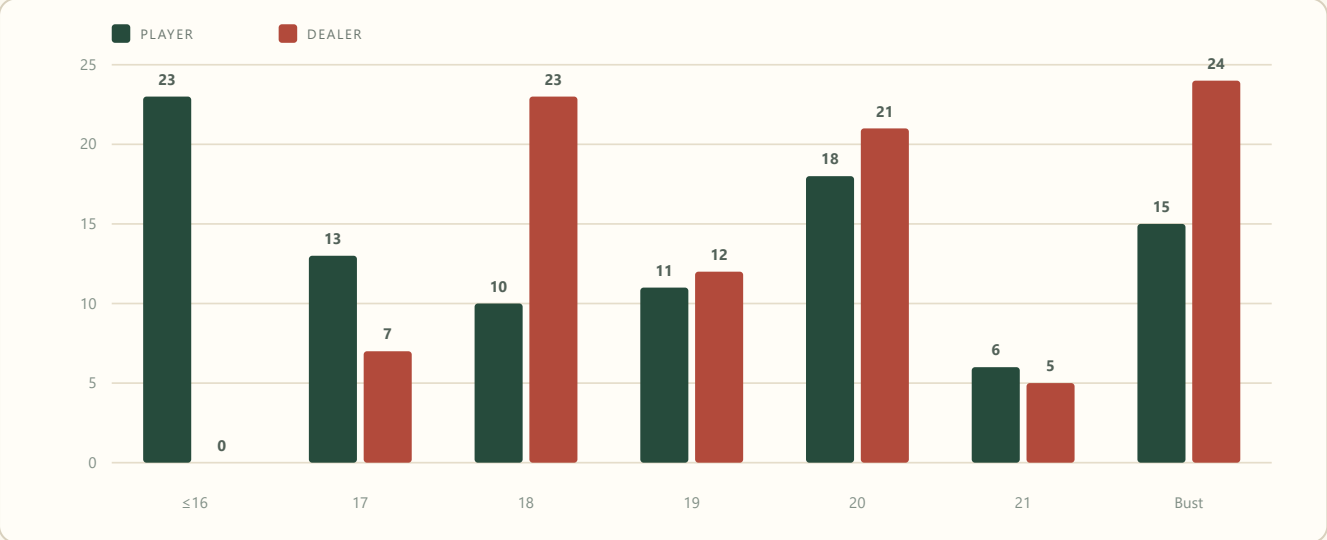
Result	This session	100,000-hand reference	Difference
Win	39 · 39.0%	43.2%	-4.2 pts
Loss	42 · 42.0%	47.5%	-5.5 pts
Push / exactly zero	19 · 19.0%	9.2%	+9.8 pts

THE PUSH PROBLEM

19 zero-net rounds is about **3.4 standard deviations** above the reference push rate. Eighteen were genuine total pushes; the nineteenth was the insurance save on a dealer blackjack. More pushes means fewer chances to win or lose — and a flatter bankroll.

You vs the CPU dealer

Played-out rounds only: the 8 rounds that ended on a natural or a dealer blackjack are excluded, leaving 92 dealer totals and 96 player hands.



Dealer 18 is a wall. The dealer made 18 23 times, more than any other total; the player made 18 only 10 times.

The dealer busted 24 times; the player busted 15 hands. Standing pat is not passive here — it is the whole game.

The scorecard

Every round, in order. Net is in units of the flat one-unit bet. EV is the coach's expectation for the first graded action; a dash means the round ended before a graded decision.

#	YOU	DEALER	PLAY	NET	EV	✓
01	10♥ A♦	9♥ J♦	—	W +1.50	—	—
02	K♦ 3♣ 9♦	K♦ 6♥ K♣	H	L -1.00	-0.43	✓
03	2♥ K♣ 2♥ 5♣	8♠ 4♥ K♣	H · H · S	W +1.00	-0.27	✓
04	7♦ J♠	5♦ 6♦ 8♦	S	L -1.00	-0.04	✓
05	8♠ Q♦	K♦ 5♠ 3♣	S	P +0.00	—	✓
06	3♥ J♥ 8♦	J♣ J♦	H	W +1.00	-0.43	✓
07	5♥ 10♠ 7♦	8♦ 7♣ 7♣	H	L -1.00	-0.42	✓
08	5♥ 9♣	6♥ J♥ 10♦	S	W +1.00	-0.15	✓
09	Q♥ 2♥	6♥ 3♥ 9♠	S	L -1.00	-0.15	✓
10	K♠ 8♠	K♣ 7♥	S	W +1.00	—	✓
11	A♦ 8♦	3♦ J♦ 5♣	S	W +1.00	+0.40	✓
12	Q♣ 6♦	6♠ 7♥ 3♠ 6♠	S	W +1.00	-0.15	✓
13	5♣ 3♠ 9♦	A♣ 9♠	I · H · S	L -1.00	-0.20	✓
14	8♣ 4♥ 5♣	K♠ 7♠	H · S	P +0.00	-0.38	✓
15	J♣ J♣	5♦ J♥ 3♦	S	W +1.00	+0.67	✓
16	2♣ 8♣ 9♥	4♦ K♣ K♥	D	W +2.00	+0.46	✓
17	A♠ J♥	8♠ J♣	—	W +1.50	—	—
18	9♣ 6♥	4♣ 7♣ K♠	S	L -1.00	-0.21	✓
19	K♥ 5♦	J♥ A♥	—	L -1.00	—	—
20	Q♠ 9♣	5♦ 7♥ 7♣	S	P +0.00	—	✓
21	3♥ 3♦ Q♣ 8♠	8♠ K♣	H · H	L -1.00	-0.22	✓
22	4♦ 8♠ Q♦	9♥ 2♠ 7♣	H	L -1.00	-0.34	✓
23	A♥ 4♦ / A♦ K♠	K♠ 8♠	P	P +0.00	+0.18	✓
24	5♠ 8♥	Q♠ A♣	—	L -1.00	—	—
25	K♠ 5♦	6♣ 10♠ K♦	S	W +1.00	-0.15	✓
26	Q♦ 6♦	4♦ J♣ K♥	S	W +1.00	-0.21	✓
27	6♣ A♣ 5♣	3♠ A♥ A♥ A♦ 4♠	D	L -2.00	+0.06	✓
28	10♣ 3♣ 10♦	8♦ 2♣ J♦	H	L -1.00	-0.32	✓
29	10♥ K♦	2♣ 3♠ 9♠ 9♥	S	W +1.00	+0.64	✓
30	7♣ A♣ 8♣ 5♣	A♦ 2♣ 2♥ 9♠ A♦ 5♥	I · H · H	W +1.00	-0.09	✓
31	Q♦ 7♦	5♣ K♦ 6♠	S	L -1.00	-0.04	✓
32	2♣ 4♠ 9♥	3♦ K♠ 6♣	H · S	L -1.00	-0.11	✓
33	7♠ 2♦ 4♣ 3♠ 4♠	8♦ 4♥ 3♠ 6♠	H · H · H · S	L -1.00	+0.10	✓
34	J♠ 7♣	6♦ 8♠ K♣	S	W +1.00	+0.01	✓

Play: H hit · S stand · D double · P split · I insurance.

The scorecard

Every round, in order. Net is in units of the flat one-unit bet. EV is the coach's expectation for the first graded action; a dash means the round ended before a graded decision.

#	YOU	DEALER	PLAY	NET	EV	✓
35	8♣ 2♥	A♥ 10♥	I	P +0.00	+0.01	✓
36	7♣ 4♥ 7♠	2♣ 10♠ 2♠ 4♠	D	P +0.00	+0.47	✓
37	8♣ 6♣ 10♥	8♣ 10♥	H	L -1.00	-0.37	✓
38	A♥ Q♦	7♠ J♥	—	W +1.50	—	—
39	4♣ 6♦	Q♦ A♦	—	L -1.00	—	—
40	K♦ J♦	3♣ 9♥ A♦ 10♥	S	W +1.00	+0.65	✓
41	5♦ 9♣	6♣ 9♥ 7♠	S	W +1.00	-0.15	✓
42	J♠ Q♣	J♥ K♦	S	P +0.00	+0.55	✓
43	K♥ 4♦ Q♣	8♣ 4♦ 9♣	H	L -1.00	-0.37	✓
44	10♥ 8♣	8♦ K♥	S	P +0.00	—	✓
45	K♦ 6♦ 10♦	Q♠ 10♣	H	L -1.00	-0.54	✓
46	3♠ J♠	4♣ 5♠ 6♠ 5♦	S	L -1.00	-0.21	✓
47	6♦ 2♦ J♣	Q♦ K♦	H · S	L -1.00	-0.25	✓
48	9♣ J♠	5♣ J♥ 2♥	S	W +1.00	—	✓
49	J♦ 7♣	4♦ 4♦ 9♣	S	P +0.00	-0.08	✓
50	3♣ J♠	6♣ 6♠ 6♦	S	L -1.00	-0.15	✓
51	3♣ 9♥ 9♠	8♥ A♥	H	W +1.00	-0.27	✓
52	10♦ 10♥	7♣ A♦	S	W +1.00	+0.77	✓
53	9♠ 7♣ 4♠	K♣ 3♦ A♦ 6♠	H · S	P +0.00	-0.54	✓
54	10♥ 3♦ Q♣	8♦ A♥	H	L -1.00	-0.32	✓
55	7♠ 4♦ 9♣	Q♣ 9♦	D	W +2.00	+0.18	✓
56	10♠ 6♠	3♥ 8♠ 4♥ 8♣	S	W +1.00	-0.25	✓
57	A♥ 3♥ Q♥ 5♠	8♣ 6♠ 5♣	H · H · S	P +0.00	+0.01	✓
58	8♦ 10♦	Q♦ 8♣	S	P +0.00	—	✓
59	5♦ 5♠ K♥	2♠ 2♣ J♥ 9♥	D	W +2.00	+0.36	✓
60	J♥ Q♠	A♦ 9♣	I · S	P +0.00	+0.66	✓
61	3♠ J♣ K♣	A♠ 3♠ 9♠ 5♠	I · H	L -1.00	-0.40	✓
62	2♦ 10♣ 6♣ / 2♠ K♥ 7♣	7♣ 7♣ 10♠	P · H · S · H · S	W +2.00	+0.01	✓
63	A♥ 5♥ 10♣	4♠ 4♣ J♠	D	L -2.00	+0.06	✓
64	Q♣ 6♦ 2♦	10♥ 8♦	H · S	P +0.00	-0.54	✓
65	Q♠ 10♦	Q♦ Q♦	S	P +0.00	+0.55	✓
66	A♦ 9♥	Q♦ 4♣ 7♠	S	L -1.00	+0.55	✓
67	4♠ 6♦ 2♥	8♠ 8♣ 7♠	D	W +2.00	+0.29	✓

The scorecard

Every round, in order. Net is in units of the flat one-unit bet. EV is the coach's expectation for the first graded action; a dash means the round ended before a graded decision.

#	YOU	DEALER	PLAY	NET	EV	✓
68	8♦ 6♠	2♠ A♦ 6♠	S	L -1.00	-0.29	✓
69	K♥ 6♦	2♣ A♣ 2♦ Q♣ 5♣	S	L -1.00	-0.29	✓
70	K♠ J♥	7♥ 3♠ 5♣ J♦	S	W +1.00	+0.77	✓
71	10♦ 4♣ A♠ 7♣	8♠ 7♦ 7♥	H · H	L -1.00	-0.37	✓
72	8♠ A♣	9♥ 9♥	S	W +1.00	+0.29	✓
73	J♦ 5♠ 2♣	K♥ J♣	H · S	L -1.00	-0.50	✓
74	Q♦ 5♠ Q♠	9♦ 10♣	H	L -1.00	-0.47	✓
75	K♠ J♥	4♠ 6♦ J♦	S	P +0.00	+0.66	✓
76	A♥ 10♥ / A♣ 3♣	5♠ 10♦ 10♠	P	W +2.00	+0.61	✓
77	2♥ A♠ J♠	4♥ 6♥ 2♥ 2♥ 10♥	H · S	W +1.00	+0.10	✓
78	J♥ 8♠	8♥ 7♣ 10♠	S	W +1.00	—	✓
79	5♣ 4♠ 2♦ 3♠ 7♥	7♥ A♠	H · H · H	W +1.00	+0.17	✓
80	K♥ 3♦	5♠ 7♥ 4♣ A♣	S	L -1.00	-0.17	✓
81	Q♦ 5♥	6♥ J♦ Q♠	S	W +1.00	-0.15	✓
82	K♣ 3♠ 9♦	8♣ Q♠	H	L -1.00	-0.32	✓
83	J♦ 4♥ J♥	7♠ 5♣ 7♥	H	L -1.00	-0.32	✓
84	10♠ J♠	8♣ 7♦ 4♠	S	W +1.00	+0.79	✓
85	3♦ 4♥ 2♣ 5♦ A♥ 2♣	J♦ J♣	H · H · H · H · S	L -1.00	-0.32	✓
86	J♥ 7♣	J♥ 10♣	S	L -1.00	-0.42	✓
87	7♥ K♠	9♥ 6♥ 3♣	S	L -1.00	-0.42	✓
88	7♣ J♣ / 7♠ K♦	4♣ 6♣ 8♣	P · S · S	L -2.00	+0.05	✓
89	2♥ Q♠ 5♠	K♦ Q♣	H · S	L -1.00	-0.38	✓
90	10♠ K♦	Q♥ Q♣	S	P +0.00	+0.55	✓
91	A♣ K♣	4♣ K♣	—	W +1.50	—	—
92	5♣ 5♥ 5♥	3♦ 4♦ 2♣ A♥	D	L -2.00	+0.41	✓
93	4♠ 10♥ 5♠	K♥ 3♥ K♠	H · S	W +1.00	-0.47	✓
94	4♠ A♠ 9♥	6♣ A♥	D	L -2.00	+0.18	✓
95	A♦ 5♣ 2♠	7♥ 10♥	H · S	W +1.00	-0.00	✓
96	7♦ 3♣ 10♦	6♣ 4♦ K♥	D	P +0.00	+0.58	✓
97	9♠ 5♦ 5♣	A♥ 7♠	I · H · S	W +1.00	-0.44	✓
98	Q♦ 10♣	2♦ 5♠ 5♠ 2♣ J♣	S	W +1.00	+0.64	✓
99	9♠ K♥	3♠ 10♣ 6♣	S	P +0.00	—	✓
100	6♥ Q♥ K♣	K♦ 8♣	H	L -1.00	-0.54	✓

HOW THIS WAS PLAYED

Reproducible, *not prophetic.*

This is a headless session through the same `PlayGame` engine that powers the playable table, not a hand-typed sequence. The seed fixes every shuffle and every draw.

Rules

Decks	6
Dealer	Stands on soft 17
Blackjack	3:2
Double	Any two · after split
Splits	Up to 4 hands · aces one card
Surrender	None
Dealer peek	US peek
Insurance	Offered

Session

Seed	20261029
Bankroll	200.00 units
Bet	1 unit, flat
Penetration	75%
Strategy	Total-dependent chart
Insurance	Take when ten density makes EV ≥ 0
Grading	Chart EV for the chosen action
Code	6e0fc28

What is exact

Dealer distributions and action EVs are exact for their composition. The 0.426% fresh-shoe edge is the composition-aware fixed-strategy solve; its split EV uses the infinite-deck approximation below.

The 100,000-hand reference uses the same play engine and policy. Its win/loss/push rates and 1.153-unit standard deviation are the long-run baseline used on the previous pages.

What is approximated

Split EV uses the documented independent-hand approximation. The chart is total-dependent, not composition-dependent. The insurance grade uses the unseen ten density with the hole card treated as one unseen card, not a peek.

The play engine settles the real shoe composition, but the coach's EV is chart-based.

What this does not prove

One hundred rounds cannot distinguish a 0.426% edge from zero. The coach ledger is not the whole table: 85 graded rounds expected -1.21 and returned -6.00; the other 15 settled before a grade and returned +6.00. Those pieces reconcile to +0.00.

REPRODUCE IT

```
node reports/build-blackjack-report.mjs
```

The script writes the session JSON, this HTML, and the PDF. The reference seed is 8675309; the session seed is 20261029.

THE LAST WORD

One hundred hands, zero mistakes, zero units. The chart did its job; the cards simply refused to make a story out of it.